I focused the analysis on the primary indicator: the **Maternal Mortality Ratio (MMR) per 100,000 live births**, specifically analyzing the trends from the year **2000 to 2020** and the differences across **Income Groups**.
The output charts and figures generated from the analysis are displayed at the top of this page. Below are the findings and the statistical software output tables formatted for your appendix.

1. Descriptive Statistics

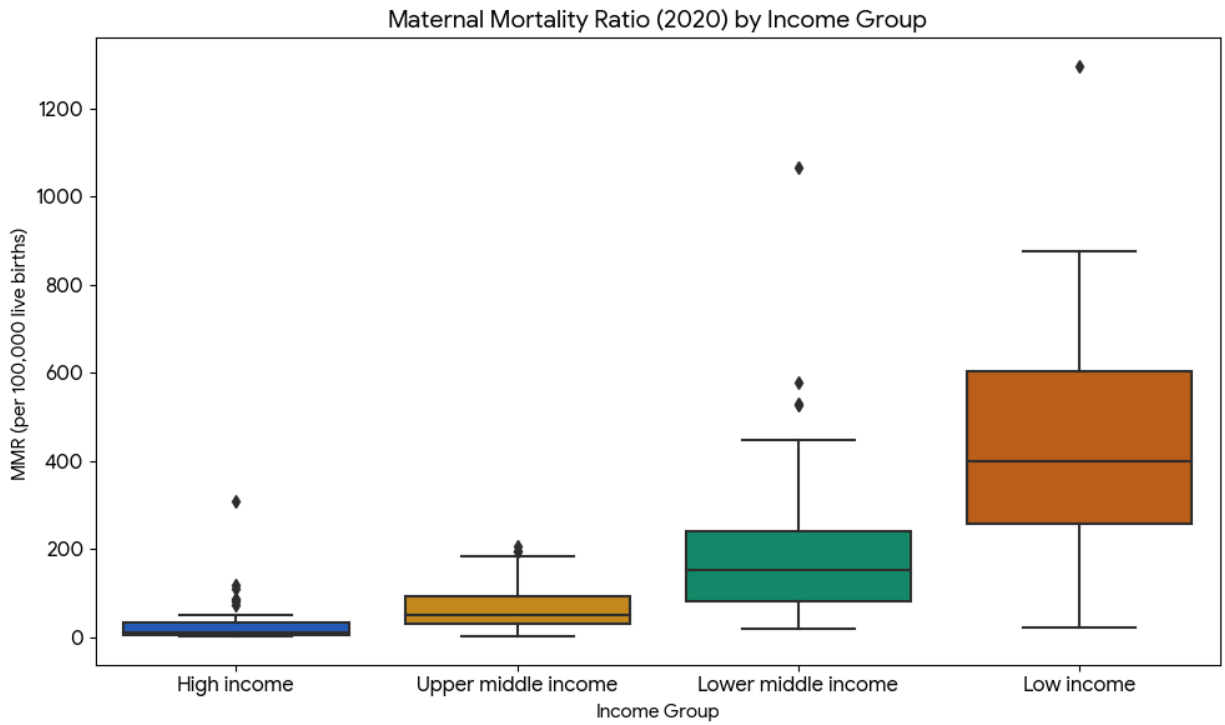
Income Group Distribution:
A frequency analysis was conducted on the country income groups. Out of the 215 valid countries/territories, the largest proportion falls into the "High income" category (40.0%), followed by "Upper middle income" (25.1%), "Lower middle income" (23.3%), and "Low income" (11.6%).
Maternal Mortality Trends (2000 vs. 2020):
The average Maternal Mortality Ratio (MMR) across all reporting countries (N=194) saw a substantial decrease over two decades. In the year 2000, the global average was **235.6** maternal deaths per 100,000 live births (SD = 332.4). By 2020, this average nearly halved to **134.9** deaths per 100,000 live births (SD = 194.3).

2. Inferential Tests

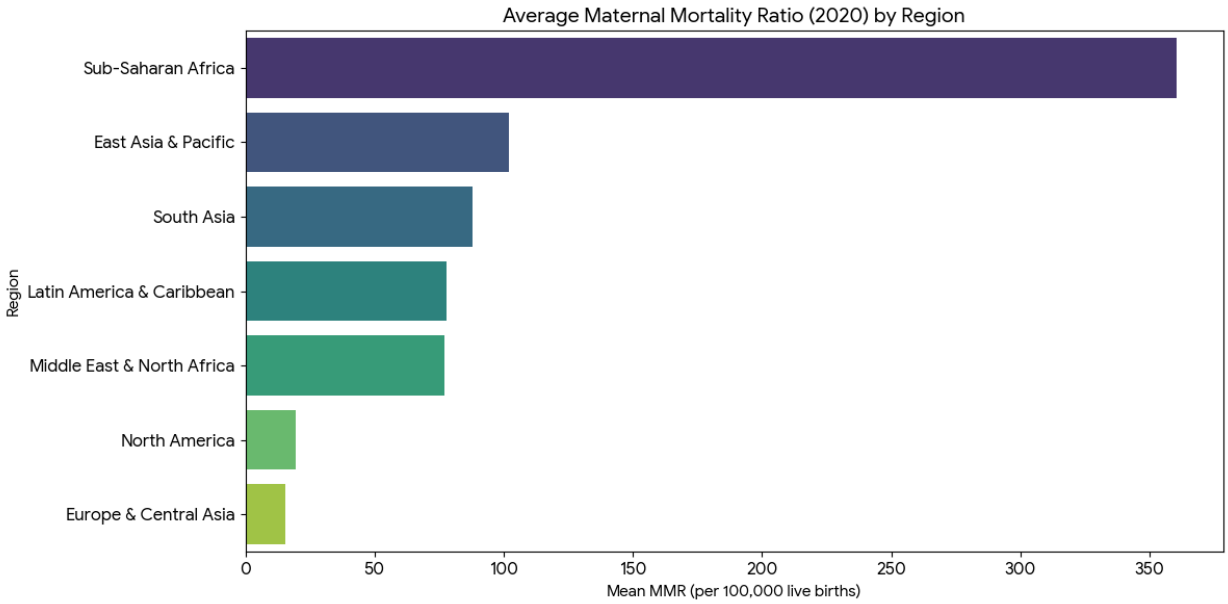
Paired Samples T-Test:
To determine if the decline in maternal mortality over the 20-year period was statistically significant, a paired samples t-test was conducted. The results indicated a highly significant decrease in MMR from 2000 to 2020 ($t(193) = 7.55, p < .001$).
One-Way ANOVA:
A One-Way Analysis of Variance (ANOVA) was conducted to assess if there are significant differences in the 2020 Maternal Mortality Ratio based on a country's income group. The test revealed a statistically significant difference across the four income levels ($F(3, 188) = 55.36, p < .001$). High-income countries exhibited drastically lower mortality ratios compared to lower-income nations, as visibly supported by the generated boxplot.

3. Visualizations

- **Figure 1: mmr_boxplot_income** - A boxplot visualizing the distribution of the 2020 Maternal Mortality Ratio across different income classifications. It clearly shows the right-skewed distribution and higher variance in low-income countries.



- **Figure 2: mmr_barplot_region.png** - A bar chart depicting the average Maternal Mortality Ratio by geographic region in 2020, highlighting disparities (e.g., Sub-Saharan Africa having the highest averages).



Appendix: Output Tables

Table 1: Frequencies - Income Group

Income Group	Frequency	Percent (%)	Valid Percent (%)	Cumulative Percent (%)
High income	86	39.6	40.0	40.0
Upper middle income	54	24.9	25.1	65.1
Lower middle income	50	23.0	23.3	88.4
Low income	25	11.5	11.6	100.0
Total Valid	**215**	**99.0**	**100.0**	
Missing	2	1.0		
Total	**217**	**100.0**		

Table 2: Descriptive Statistics

Indicator (MMR per 100k)	N	Minimum	Maximum	Mean	Std. Deviation
Year 2000	194	4.0	1658.0	235.60	332.405
Year 2020	194	2.0	1296.0	134.88	194.261
Valid N (listwise)	194				

Table 3: Paired Samples Test (Year 2000 vs. 2020)

	Mean Difference	Std. Deviation	t	df	Sig. (2-tailed)

Pair 1: MMR 2000 - MMR 2020	100.72	185.87	7.553	193	< .001

Table 4: ANOVA - Maternal Mortality Ratio (2020) by Income Group

	Sum of Squares	**df**	**Mean Square**	**F**	**Sig.**
Between Groups	3408171.0	3	1136057.0	55.362	< .001
Within Groups	3857888.0	188	20520.6		
Total	**7266059.0**	**191**			